6 (a)	The alternating current causes a **change in the magnetic flux density** in the solenoid. This causes a **change in the magnetic flux linkage in coil X**. According to **Faraday's law**, when there is a change in the magnetic flux linkage in coil X, an e.m.f. will be induced in the coil.	**B1** **B1**
(b)	$B_{max} = u_0 nI$ $= (4\pi \times 10^{-7})(2500)(2.6)$ $= 8.1681 \times 10^{-3}$ T $= \underline{8.2 \times 10^{-3}\ \text{T}}$ (2 sf)	 **A1**
(c)	$\Phi_{max} = NBA$ $= (100)(8.1681 \times 10^{-3})(7.4 \times 10^{-5})$ $= \underline{6.0 \times 10^{-5}\ \text{Wb-turns}}$ (2 sf)	 **C1** **A1**
(d)	induced e.m.f. $= -d\Phi/dt$ $= -d(NBA)/dt$ $= -d(\Phi_{max} \sin 6\pi t)/dt$ $= 6\pi(\Phi_{max}) \cos 6\pi t$ Max e.m.f inducted $= 6\pi\ (6.0 \times 10^{-5})$ $= 1.13 \times 10^{-3}$ V $= \underline{1.1 \times 10^{-3}\ \text{V}}$ (2 sf)	 **C1** **A1**